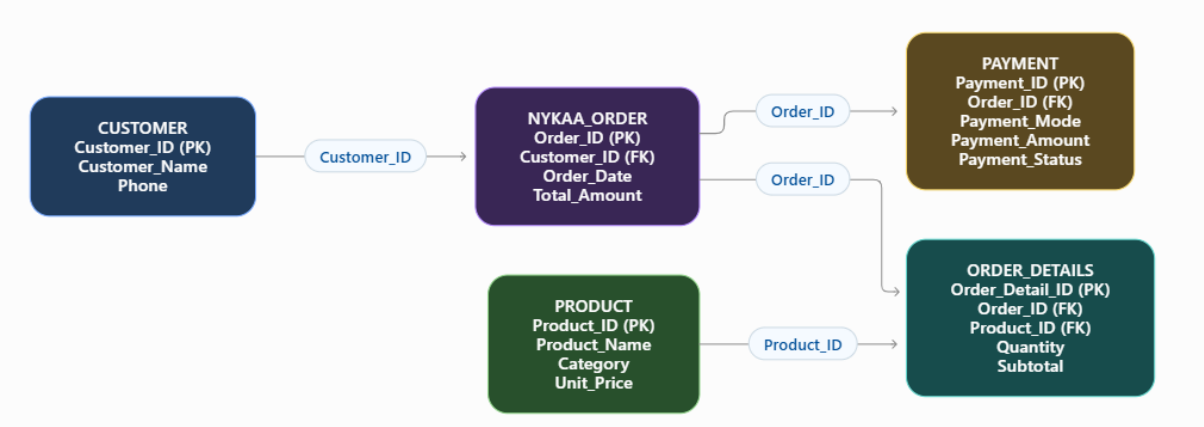
NYKAA BEAUTY STORE MANAGEMENT SYSTEM

DATABASE RELATIONSHIP ANALYSIS USING JOINS

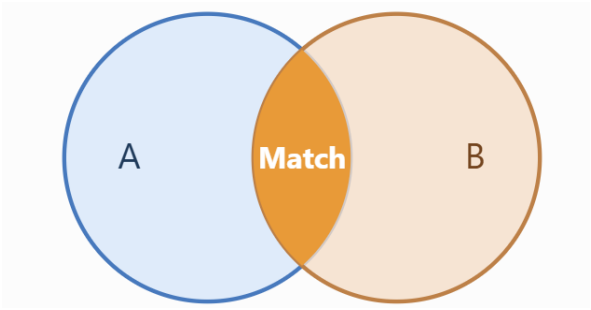
1. COMBINING MULTIPLE TABLES

Definition: JOIN operations are used to combine related information from multiple tables.



2. INNER JOIN

Definition: INNER JOIN returns only records that have matching values in both tables.



2.1 Customer and Orders

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.Customer_Name,
  4      o.Order_ID,
  5      o.Order_Date,
  6      o.Total_Amount
  7  FROM Customer c
  8  INNER JOIN Orders o
  9  ON c.Customer_ID = o.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
          1 Anitha Kumar                                          101
16-SEP-26         2700

          2 Rahul                                                 102
15-SEP-26         1800

          2 Rahul                                                 103
17-SEP-26         1200
```

2.2 Orders and Payment

```
SQL> SELECT
  2      o.Order_ID,
  3      o.Order_Date,
  4      o.Total_Amount,
  5      p.Payment_Mode,
  6      p.Payment_Amount,
  7      p.Payment_Status
  8  FROM Orders o
  9  INNER JOIN Payment p
 10  ON o.Order_ID = p.Order_ID;

  ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE                    PAYMENT_AMOUNT
---------- --------- ------------ ------------------------------ --------------
PAYMENT_STATUS
--------------------
       101 16-SEP-26         2700 UPI                                      2700
Paid

       102 15-SEP-26         1800 Credit Card                              1800
Paid
```

2.3 Product and Order_Details

```
SQL> SELECT
  2      p.Product_ID,
  3      p.Product_Name,
  4      od.Order_ID,
  5      od.Quantity,
  6      od.Unit_Price,
  7      od.Subtotal
  8  FROM Product p
  9  INNER JOIN Order_Details od
 10  ON p.Product_ID = od.Product_ID;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  ORDER_ID   QUANTITY UNIT_PRICE   SUBTOTAL
---------- ---------- ---------- ----------
       201
Organic Turmeric Powder
       101          2       1000       2000

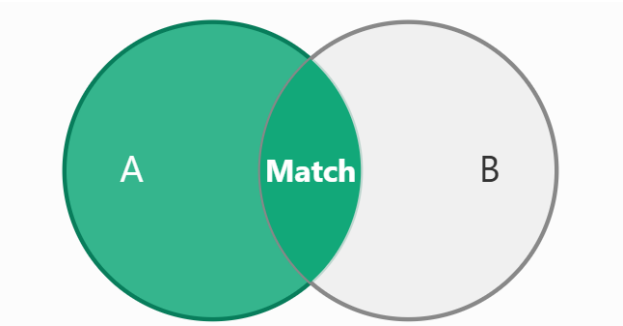
       202
Black Pepper Face Wash
       101          1        700        700

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  ORDER_ID   QUANTITY UNIT_PRICE   SUBTOTAL
---------- ---------- ---------- ----------

       202
Black Pepper Face Wash
```

3. LEFT JOIN

Definition: LEFT JOIN returns all records from the left table and matching records from the right table. If no match exists, the right-table columns show NULL.



3.1 Display All Customers and Their Orders

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.Customer_Name,
  4      o.Order_ID,
  5      o.Order_Date,
  6      o.Total_Amount
  7  FROM Customer c
  8  LEFT JOIN Orders o
  9  ON c.Customer_ID = o.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
          1 Anitha Kumar                                            101
16-SEP-26         2700

          2 Rahul                                                   102
15-SEP-26         1800

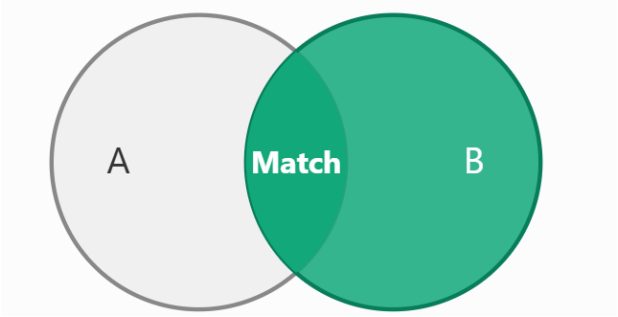
          2 Rahul                                                   103
17-SEP-26         1200


CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
          4 Divya


          3 Priya
```

4. RIGHT JOIN

Definition: RIGHT JOIN returns all records from the right table and matching records from the left table.



4.1 Display All Orders and Customer Details

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.Customer_Name,
  4      o.Order_ID,
  5      o.Order_Date,
  6      o.Total_Amount
  7  FROM Customer c
  8  RIGHT JOIN Orders o
  9  ON c.Customer_ID = o.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
          1 Anitha Kumar                                          101
16-SEP-26         2700

          2 Rahul                                                 102
15-SEP-26         1800

          2 Rahul                                                 103
17-SEP-26         1200
```

5. COMPLETE ORDER DETAILS

This query combines Customer, Orders, Order_Details, Product and Payment tables.

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.Customer_Name,
  4      o.Order_ID,
  5      o.Order_Date,
  6      p.Product_ID,
  7      p.Product_Name,
  8      od.Quantity,
  9      od.Unit_Price,
 10      od.Subtotal,
 11      py.Payment_Mode,
 12      py.Payment_Amount,
 13      py.Payment_Status
 14  FROM Customer c
 15  INNER JOIN Orders o
 16      ON c.Customer_ID = o.Customer_ID
 17  INNER JOIN Order_Details od
 18      ON o.Order_ID = od.Order_ID
 19  INNER JOIN Product p
 20      ON od.Product_ID = p.Product_ID
 21  INNER JOIN Payment py
 22      ON o.Order_ID = py.Order_ID
 23  ORDER BY o.Order_Date DESC;

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT PRODUCT_ID
--------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE   SUBTOTAL PAYMENT_MODE                  PAYMENT_AMOUNT
---------- ---------- ---------- ------------------------------ --------------
PAYMENT_STATUS
--------------------
          1 Anitha Kumar                                             101
```

6. CUSTOMER PURCHASE HISTORY

This query displays each customer's orders and the products purchased.

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.Customer_Name,
  4      o.Order_ID,
  5      o.Order_Date,
  6      p.Product_Name,
  7      od.Quantity,
  8      od.Subtotal
  9  FROM Customer c
 10  INNER JOIN Orders o
 11      ON c.Customer_ID = o.Customer_ID
 12  INNER JOIN Order_Details od
 13      ON o.Order_ID = od.Order_ID
 14  INNER JOIN Product p
 15      ON od.Product_ID = p.Product_ID
 16  ORDER BY c.Customer_ID, o.Order_Date;

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT
---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY   SUBTOTAL
---------- ----------
          1 Anitha Kumar                                           101
16-SEP-26
Organic Turmeric Powder
         2       2000


CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT
---------
```

7. CUSTOMER PURCHASE HISTORY WITH PAYMENT

```
SQL> SELECT
  2      c.Customer_Name,
  3      o.Order_ID,
  4      o.Order_Date,
  5      p.Product_Name,
  6      od.Quantity,
  7      od.Subtotal,
  8      py.Payment_Mode,
  9      py.Payment_Amount,
 10      py.Payment_Status
 11  FROM Customer c
 12  INNER JOIN Orders o
 13      ON c.Customer_ID = o.Customer_ID
 14  INNER JOIN Order_Details od
 15      ON o.Order_ID = od.Order_ID
 16  INNER JOIN Product p
 17      ON od.Product_ID = p.Product_ID
 18  INNER JOIN Payment py
 19      ON o.Order_ID = py.Order_ID
 20  ORDER BY c.Customer_Name, o.Order_Date;

CUSTOMER_NAME                                      ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY   SUBTOTAL PAYMENT_MODE                  PAYMENT_AMOUNT
---------- ---------- ------------------------------ --------------
PAYMENT_STATUS
--------------------
Anitha Kumar                                           101 16-SEP-26
Organic Turmeric Powder
         2       2000 UPI                                  2700
Paid
```

8. MULTI-TABLE BUSINESS REPORT

This query summarizes the number of orders, quantity purchased and total product sales for each customer.

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.Customer_Name,
  4      COUNT(DISTINCT o.Order_ID) AS Total_Orders,
  5      NVL(SUM(od.Quantity), 0) AS Total_Quantity,
  6      NVL(SUM(od.Subtotal), 0) AS Total_Sales
  7  FROM Customer c
  8  LEFT JOIN Orders o
  9      ON c.Customer_ID = o.Customer_ID
 10  LEFT JOIN Order_Details od
 11      ON o.Order_ID = od.Order_ID
 12  GROUP BY c.Customer_ID, c.Customer_Name
 13  ORDER BY Total_Sales DESC;

CUSTOMER_ID CUSTOMER_NAME                                          TOTAL_ORDERS
----------- ---------------------------------------------------- ------------
TOTAL_QUANTITY TOTAL_SALES
-------------- -----------
          2 Rahul                                                             2
             4        3000

          1 Anitha Kumar                                                      1
             3        2700

          3 Priya                                                             0
             0           0


CUSTOMER_ID CUSTOMER_NAME                                          TOTAL_ORDERS
----------- ---------------------------------------------------- ------------
TOTAL_QUANTITY TOTAL_SALES
-------------- -----------
          4 Divya                                                             0
             0           0
```

9. PAYMENT TRANSACTION REPORT

```
SQL> SELECT
  2      py.Payment_ID,
  3      c.Customer_Name,
  4      o.Order_ID,
  5      o.Order_Date,
  6      py.Payment_Mode,
  7      py.Payment_Amount,
  8      py.Payment_Status
  9  FROM Payment py
 10  INNER JOIN Orders o
 11      ON py.Order_ID = o.Order_ID
 12  INNER JOIN Customer c
 13      ON o.Customer_ID = c.Customer_ID
 14  ORDER BY py.Payment_ID;

PAYMENT_ID CUSTOMER_NAME                                      ORDER_ID
---------- -------------------------------------------------- ----------
ORDER_DAT PAYMENT_MODE                    PAYMENT_AMOUNT PAYMENT_STATUS
--------- ------------------------------ -------------- --------------------

       301 Anitha Kumar                                          101
16-SEP-26 UPI                                    2700 Paid

       302 Rahul                                                 102
15-SEP-26 Credit Card                            1800 Paid
```